# Much Ado about Prompting

*Text Classification in Experimental Economics*

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# Eliciting subjects' minds, with text

*Methods for eliciting reasoning in text.*

| | |
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| *Cooper & Kagel* (2005) | team chat |
| *Charness & Dufwenberg* (2006) | open-ended messages |
| *Burchardi & Penczynski* (2014) | team communication protocol |
| *Capra* (2019) | think-aloud protocol |
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| *Arad, Hugh-Jones & Penczynski* (2024) | chat among subjects |
| *Arad & Severin* (2026) | instructions to an LLM |

# Classifying subjects' text

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| *Unsupervised ML* | Hanaki & Ozkes 2023; Andres et al. 2023 |

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| *LLMs* | this paper |

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# Text as data, beyond experimental economics

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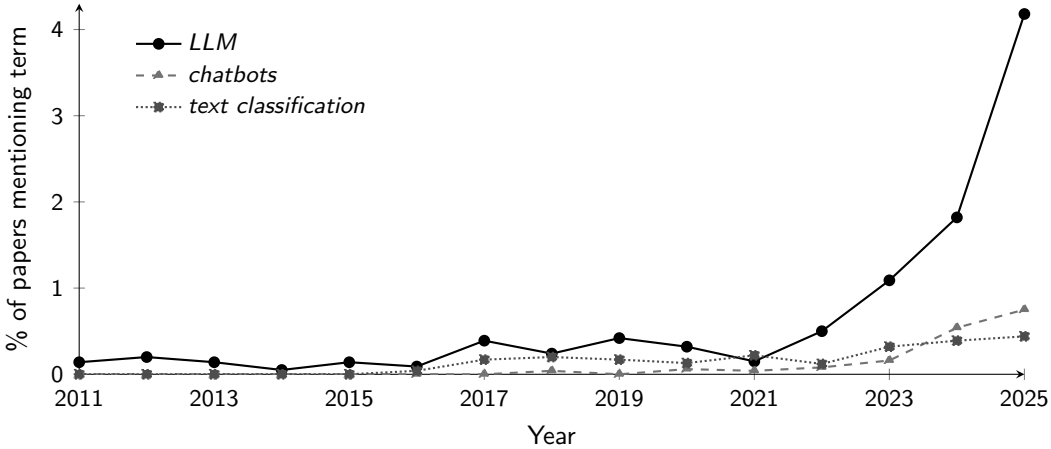
| *Qualitative interviews with AI* | Chopra & Haaland 2024; Velez & Liu 2024; Geiecke & Jaravel 2025 |

# Interest in top economics journals




14 journals, ∼48,000 papers, 2011–2025. Source: Gentzkow's econlit pipeline, `paulgp.com/econlit-pipeline`.

# Why LLMs change the calculus

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| *Faster* | $\sim$1 minute per 1,000 items |
| *Stronger* | complex concepts — promises, level-$k$ reasoning |

# Before LLMs: codebooks for humans

To classify experimental text, a researcher writes a *codebook* to train research assistants. It typically contains:

| | |
|---|---|
| *Experiment context* | game mechanics, decision environment |
| *Theory* | the framework behind the categories |
| *Category definitions* | inclusion / exclusion criteria per label |
| *Annotated examples* | walk-through to cover edge cases |

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Can a researcher prepare a codebook for an *LLM* the same way they would for a *human annotator*?

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*And for the referee:* how should a study's prompt be evaluated to judge whether the classifications can be trusted?

## “How should I prompt my model?”

*Three strands of literature greet the researcher.*

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3. *Prompt brittleness*

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| *Use the system message; examples as dialogue* | Törnberg 24 |
| *Follow the vendor's prompting guide* | OpenAI, Anthropic, Google, Meta |

# Empirical results

| *Formatting matters* | *Formatting doesn't* |
| --- | --- |
| Sclar 24; Han 25 | Pangakis 23 |

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Rückdeschel 25

*Reasoning hurts*

Wu 23; Sprague 24

# Brittle prompts

| | |
|---|---|
| *"Minor wording alterations [. . . ] produce varying outputs."* | Reiß 2023 |
| *"Seemingly irrelevant formatting choices produce up to 76 pp accuracy swings."* | Sclar et al. 2024 |
| *"Rankings reverse across equivalent paraphrased instructions."* | Mizrahi et al. 2024 |
| *"Proprietary models require explicit justification of prompts because they are so sensitive to phrasing."* | Palmer & Spirling 2024 |

# The state of prompting in social science

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**The researcher's options:**

(a) *Avoid LLMs altogether.*

(b) *Wing it — and hope for the best.*

(c) *Pick advice from the pile — untested, perhaps misleading.*

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- *Guide referees on what to scrutinise.*

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- *1,182 treatments* — model $\times$ variant $\times$ task.
- *703,253 API calls*, 99.94 % successful, over 13 days (24 March — 6 April 2026).

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**RQ4.** *How does model scale affect classification performance and moderate the sensitivity to prompt content and design?*

# Preview of what we find

**Two dimensions drive everything.**

*Prompt specification* — is the prompt well-specified (detailed category definitions and worked examples) or underspecified (a one-line category name)?

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*Brittleness reflects underspecified prompts, small models, or both.*

# How we reconcile the literature

*Each study tested a different corner of specification × capability.*

| | | |
|---|---|---|
| Sclar 24: 76 pp format swings | ⇒ | underspecified prompts, small LLaMA / Mistral |
| Pangakis 23; Gilardi 23: null | ⇒ | well-specified codebooks, GPT-3.5 / 4 |
| Rückdeschel 25: detail hurts | ⇒ | small open model (Llama 8B) |
| Mizrahi 24: paraphrase flips ranks | ⇒ | small models, short task descriptions |

↗ what we find vs. these claims

# LLMs are not deterministic

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*To call an effect real, we need a paired statistical test.*

# What we mean by “accuracy”

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*Accuracy is the LLM’s agreement rate with that joint human ground truth.*

# McNemar's test

Same 2,063 items classified twice — under prompt A and prompt B.

| | *B correct* | *B wrong* |
|---|---|---|
| *A correct* | $a$ | $b$ |
| *A wrong* | $c$ | $d$ |

*Where the two agree* ($a$, $d$) tells us nothing about which prompt is better.
*Where they disagree* ($b$, $c$) is all the signal.

If A is truly better, more items should fall in $b$ (A right, B wrong) than in $c$. McNemar tests whether $b \neq c$ beyond what random flipping predicts.

*Every paired comparison in our experiments uses McNemar at $p < 0.001$.*

Accuracy estimates use 95 % Wilson confidence intervals

↗ details.

# Datasets

| | | |
|---|---|---|
| *Promise* | public goods, chat $\rightarrow$ promise / empty talk | 719 |
| *Level-k I* | voting game, team communication $\rightarrow$ level of strategic thinking 0–3 | 493 |
| *Level-k II* | coordination game, team communication $\rightarrow$ level of strategic thinking 0–5 | 851 |

*Promise*: concept familiar from pre-training. *Level-k*: domain-specific theory.

# A shared prompt template

```
# General Task -- Classify X in E

# Role Persona -- Act as a behavioural economist specialised in G

# Context
## Experiment -- game mechanics, payoffs, protocol
## Theory -- theoretical framework (L-tasks only)

# Classification Task -- categories, assignment rules

# Classification Coding -- output labels

# Examples -- worked examples (n-shot only)

# Constraints / # Output Format -- JSON
```

# Four models

| | *larger* | *smaller* |
|---|---|---|
| *Proprietary* | GPT 5.4 | GPT 5.4-nano |
| *Open weight* | Qwen 3.5 397B (17B MoE) | Qwen 3.5 9B (dense) |

| Model | \$/M input | \$/M output | vs. Qwen 9B |
|---|---|---|---|
| Qwen 3.5 9B | 0.10 | 0.15 | 1× |
| GPT 5.4-nano | 0.20 | 1.25 | ∼2.5× |
| Qwen 3.5 397B | 0.39 | 2.34 | ∼5× |
| GPT 5.4 | 2.50 | 15.00 | ∼31× |

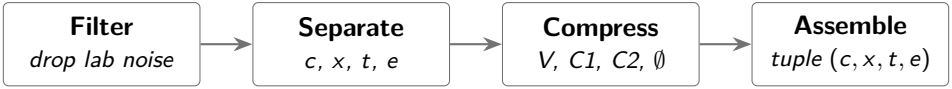
# Experiment 1 · Content ablation

*Decompose the codebook into three components; vary each independently.*

| Component | Levels |
|---|---|
| *Classification instructions* ($c$) | V, S |
| *Experiment context* ($x$) | V, C1, C2, $\emptyset$ |
| *Theory* ($t$) — $L$-tasks only | V, C1, C2, $\emptyset$ |

*Promise*: $2 \times 4 = 8$ variants. *Level-k I*: $2 \times 4 \times 4 = 32$. *Level-k II*: 32.
*72 variants $\times$ 4 models = 288 cells.*

# Experiment 1 · Codebook → prompt pipeline




$c$ = classification instructions $x$ = experimental context $t$ = theory $e$ = examples

*All LLM-assisted steps use Claude Opus 4.6.*

# Experiment 1 · Compression — how much is kept

*Iterative compression (V → C1 → C2) with a task-specific key-term list.*

| Source | Level | Cosine sim. | % words | Key terms |
|---|---|---|---|---|
| *Experiment context* | V | 1.000 | 100 | 10 |
| | C1 | 0.971 | 49 | 10/10 |
| | C2 | 0.940 | 31 | 10/10 |
| | ∅ | — | 0 | — |
| *Theory* | V | 1.000 | 100 | 9 |
| | C1 | 0.948 | 55 | 9/9 |
| | C2 | 0.839 | 26 | 9/9 |
| | ∅ | — | 0 | — |
| *Classification instructions* | V | 1.000 | 100 | — |
| | S | 0.722 | 5 | — |

*Example: Level-k II.* Cosine ↗ (what is this?) from `text-embedding-3-large`. Every key term retained at every level.

# Experiment 1 · Two information levels

**Verbatim** — the codebook, filtered but untouched. Full classification instructions ($c$), experimental context ($x$), theory ($t$).

($c_V$, $x_V$, $t_V$, $\emptyset$) — *1,000–1,500 words.*

**Minimal** — a one-line description of each category. No experimental context, no theory, no examples.

($c_S$, $\emptyset$, $\emptyset$, $\emptyset$) — *30–80 words depending on the task.*

# Experiment 2 · Format

*Same content, five visual presentations.*

| Format | md markers | titles | bullets | description |
|---|---|---|---|---|
| *ml (baseline)* | ✓ | ✓ | ✓ | `# headers` + `- bullets` |
| *tl* | | ✓ | ✓ | plain titles + bullets |
| *mp* | | ✓ | | plain titles + prose |
| *fl* | | | ✓ | no headers, bullets |
| *pt* | | | | no headers, prose |

4 alternative formats × 2 information levels (minimal, verbatim) = *8 new variants / task* × 3 tasks × 4 models = *96 cells*.

Plus two system-vs-user message splits on verbatim content.

↗ scatters · text example

# Experiment 3 · Framing

*Two framing elements, tested by removal.*

| Element | Example |
|---|---|
| *General task (GT)* | "Classify messages from a coordination game." |
| *Role persona (RP)* | "You are a behavioural economist specialised in level-$k$." |

Three removal conditions: *no RP*, *no GT*, *neither*. Compared against the full-framing baseline.
Each at both information levels (minimal, verbatim). $3 \times 3$ tasks $\times$ 2 levels = *18 comparisons / model.*

# Experiment 4 · Examples, and how to present them

**Examples.** $n$-shot from the codebook's worked examples.

$P$: 21 ($\sim$10/label) $L_1$: 13 ($\sim$3/label) $L_2$: 5 ($\sim$1/label)

Tested at *minimal* and *verbatim* content — separates content vs. examples as sources of information.

**Presentation styles.** Five:

*markdown* (baseline) · *XML tags* · *delimiter* · *arrow* · *multi-turn*

First four in the system message; *multi-turn* = user/assistant dialogue outside the system message.

↗ see the five styles

# Experiment 5 · Reasoning

*Enable model reasoning at multiple effort levels.*

| Models | Levels tested | Median tokens |
|---|---|---|
| *GPT 5.4 / nano* | low, medium, high, extra-high | 113 / 309 / 682 / 2,479 |
| *Qwen 397B / 9B* | on / off | ∼750 / 0 |

Crossed with content (verbatim / minimal) × shot (0-shot / *n*-shot) — tests whether reasoning can compensate for reduced prompt detail, and whether examples absorb reasoning gains.

# Experiment 6 · Information fidelity (1/2)

*Hold the informational content approximately constant, vary the surface expression.*

| Treatment | Length change | Cosine to verbatim |
|---|---|---|
| *Verbose* | +50 to 275 % words, relevant elaboration | 0.87–0.97 |
| *Noise* | matched length, irrelevant padding (red-squirrel facts) | 0.57–0.73 |

Each treatment is applied at 3 levels — *classification instructions only*, *experimental context only*, or *both*:

*Verbose*: 3 levels × 3 tasks = 9 comparisons / model. *Noise*: 3 levels × 3 tasks = 9 comparisons / model.

Total: *18 comparisons / model* at each shot condition (0-shot and *n*-shot).

The contrast between *verbose* and *noise* isolates *informational relevance* from *prompt length*.

↗ verbose & noise example

# Experiment 6 · Information fidelity (2/2)

Two further tests of expression-vs-content:

| Treatment | What changes |
|---|---|
| *Paraphrase* | round-trip back-translation: EN $\rightarrow$ TR $\rightarrow$ EN and EN $\rightarrow$ DE $\rightarrow$ EN<br>meaning preserved (cosine 0.97–0.99); surface wording varies |
| *Raw codebook* | original document with payment instructions, software notes, formatting artefacts — no filter, no compression |

Paraphrase: 2 paraphrases $\times$ 3 tasks $=$ 6 comparisons / model.
Raw codebook: one head-to-head against the cleaned pipeline output ($n$-shot).

# Results

# Content · the two information levels

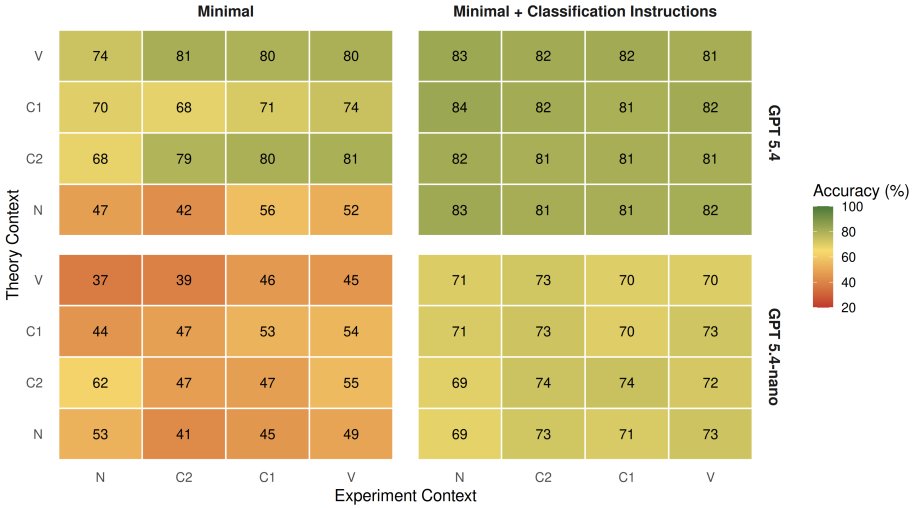
| | *Minimal* | | | *Verbatim* | | |
|---|---|---|---|---|---|---|
| | $P$ | $L_1$ | $L_2$ | $P$ | $L_1$ | $L_2$ |
| GPT 5.4 | **81.0** | 47.4 | 49.6 | **81.6** | 81.3 | **81.0** |
| GPT 5.4-nano | 72.4 | **52.5** | **53.1** | 75.0 | 70.2 | 68.9 |
| Qwen 397B | 79.1 | 50.1 | 43.6 | 76.1 | **83.0** | 74.6 |
| Qwen 9B | 74.6 | **53.1** | 38.8 | 74.1 | 70.8 | 64.4 |

*Bold* = best in each column, and any value statistically tied with it.

*Promise: done already at minimal. Level-k: jumps once content is there.*

↗ what drives this

# Content · information surface ($L_1$)




↗ 4-model surfaces

# Result 1

- *Context and theory partially substitute for missing instructions.*
- *Larger models benefit from more detailed content.*
- *Smaller models perform better with less.*

# Format & framing · proportion significant

| | *Format* | | | | *Framing* | | | |
|---|---|---|---|---|---|---|---|---|
| | Minimal | | Verbatim | | Minimal | | Verbatim | |
| | + | − | + | − | + | − | + | − |
| GPT 5.4 | **.17** | .08 | .00 | .08 | **.33** | **.11** | **.33** | .00 |
| GPT 5.4-nano | .00 | .00 | .08 | .00 | **.33** | **.22** | .00 | .00 |
| Qwen 397B | .08 | .08 | .00 | .00 | **.11** | **.33** | .00 | .00 |
| Qwen 9B | **.25** | **.17** | .00 | .00 | **.78** | .00 | .00 | **.11** |

*On well-specified prompts: almost nothing is significant. On underspecified prompts: inconsistent direction.*

↗ scatter plots · format example

# Result 2

- *Neither formatting nor framing reliably affects classification accuracy.*
- *On verbatim prompts, effects are negligible.*
- *On minimal prompts, significance increases but the direction is inconsistent across models and tasks.*

# Examples · content and examples substitute

*GPT 5.4; the pattern is similar for the other three models.*

| | $P$ | $L_1$ | $L_2$ |
|---|---|---|---|
| $\Delta_{\text{Ex}}$ | $+11.7^*$ | $+37.4^*$ | $+20.4^*$ |
| $\Delta_{\text{Ex}} - \Delta_{\text{Content}}$ | $+11.1^*$ | $+3.4$ | $-10.9^*$ |
| $\Delta_{\text{Ex} \mid \text{Content}}$ | $+9.9^*$ | $+2.6$ | $+0.7$ |
| $\Delta_{\text{Content} \mid \text{Ex}}$ | $-1.2$ | $-0.8$ | $+11.6^*$ |

$\Delta_{\text{Ex}}$: examples added to minimal. $\Delta_{\text{Ex}} - \Delta_{\text{Content}}$: examples alone vs. content alone.
$\Delta_{\text{Ex} \mid \text{Content}}$: examples given verbatim content. $\Delta_{\text{Content} \mid \text{Ex}}$: content given *n*-shot.

*Examples dominate on familiar concepts; content dominates on theory-dependent tasks.*

# Result 3

- *Content and examples carry overlapping information — partial substitutes.*
- *Examples dominate on recognition-heavy tasks.*
- *Content dominates on theory-dependent tasks.*

# Prompt $\approx$ codebook

| | $P$ | $L_1$ | $L_2$ |
|---|---|---|---|
| GPT 5.4 | **91.2** | **84.0** | **81.7** |
| GPT 5.4-nano | 79.4 | 78.3 | 70.7 |
| Qwen 397B | **90.0** | **83.6** | 77.8 |
| Qwen 9B | 84.1 | **82.8** | 68.2 |

*Verbatim + n-shot, no reasoning. Bold = best in the column, and any value statistically tied with it.*

# Example format

| | *In-prompt formats* | | | | *Multi-turn* | | | |
|---|---|---|---|---|---|---|---|---|
| | Minimal | | Verbatim | | Minimal | | Verbatim | |
| | + | − | + | − | + | − | + | − |
| GPT 5.4 | .00 | .00 | .00 | .00 | .00 | **.33** | .00 | **.33** |
| GPT 5.4-nano | .00 | .00 | .00 | .00 | .00 | **.67** | .00 | .00 |
| Qwen 397B | .00 | **.33** | .00 | **.33** | .00 | **.67** | .00 | **.33** |
| Qwen 9B | **.22** | **.33** | .00 | **.33** | .00 | **.67** | .00 | **.33** |

*In-prompt format: indifferent. Examples outside the system message: consistently hurts.*

↗ the five example styles

# Result 4

- *How examples are formatted within the system prompt does not matter.*
- *Placing examples outside the system prompt (as multi-turn conversation) consistently hurts all models.*

# Reasoning · can it substitute for content?

*GPT 5.4; the pattern is similar for the other three models.*

| | $P$ | $L_1$ | $L_2$ |
|---|---|---|---|
| $\Delta_{\text{Rea}}$ | $+6.6^*$ | $-11.5^*$ | $+19.7^*$ |
| $\Delta_{\text{Rea}} - \Delta_{\text{Content}}$ | $+5.9^*$ | $-45.4^*$ | $-11.6^*$ |
| $\Delta_{\text{Content} \mid \text{Rea}}$ | $-6.8^*$ | $+46.7^*$ | $+11.6^*$ |
| $\Delta_{\text{Rea} \mid \text{Content}}$ | $-0.8$ | $+1.2$ | $+0.0$ |

$\Delta_{\text{Rea}}$: reasoning added to minimal. $\Delta_{\text{Rea}} - \Delta_{\text{Content}}$: reasoning alone vs. content alone.
$\Delta_{\text{Content} \mid \text{Rea}}$: content given reasoning is on. $\Delta_{\text{Rea} \mid \text{Content}}$: reasoning given content is on.

*Content keeps its full value when reasoning is on; reasoning adds nothing once content is present.*

↗ 0-shot vs. *n*-shot

# Result 5

- *With codebook content in the prompt, reasoning is redundant for frontier models — and adding examples only deepens that redundancy.*
- *Smaller models gain from reasoning on recognition-heavy tasks, but those gains too are nullified once examples are present.*
- *Reasoning cannot substitute for codebook content.*
- *On underspecified prompts, reasoning can even backfire.*

# Cost of reasoning

| Reasoning effort | Cost ratio | Latency |
|---|---|---|
| *Low* | 0.38× prompt | 2.5× |
| *Medium* | 1.09× prompt | 4.7× |
| *High* | 2.53× prompt | 8.1× |
| *Extra-high* | 8.42× prompt | 21× |

*GPT 5.4 Azure pricing, April 2026.*

# Information fidelity · verbose and noise

| | *0-shot* | | | | *n-shot* | | | |
|---|---|---|---|---|---|---|---|---|
| | Verbose | | Noise | | Verbose | | Noise | |
| | + | − | + | − | + | − | + | − |
| GPT 5.4 | .00 | .00 | .00 | **.11** | .00 | .00 | .00 | .00 |
| GPT 5.4-nano | **.11** | .00 | .00 | **.11** | .00 | .00 | .00 | **.11** |
| Qwen 397B | **.44** | .00 | **.22** | .00 | .00 | **.22** | .00 | **.33** |
| Qwen 9B | **.44** | .00 | .00 | **.11** | .00 | **.11** | .00 | **.33** |

*Proportion of statistically significant comparisons vs. the verbatim baseline. 9 per cell (3 scopes × 3 tasks).*

*GPT insensitive. Qwen responsive at 0-shot; at n-shot the verbose gains disappear — examples absorb the extra information.*

↗ mean $\bar{\Delta}$ & $\sigma$ · figures · example text

# Information fidelity · paraphrase and raw codebook

| | *Paraphrase* (sig/6) | | *Raw codebook* |
|---|---|---|---|
| | 0-shot | *n*-shot | (mean $\bar{\Delta}$, sig/3) |
| GPT 5.4 | 1/6 | 0/6 | −0.5 pp, 0/3 |
| GPT 5.4-nano | 0/6 | **2/6** | **−3.0 pp, 2/3** |
| Qwen 397B | 1/6 | **2/6** | **−2.8 pp, 2/3** |
| Qwen 9B | **2/6** | **2/6** | **−6.8 pp, 2/3** |

Paraphrase: 6 comparisons / model per shot condition (2 paraphrases × 3 tasks). Median $|\bar{\Delta}| < 1$ pp throughout.
Raw codebook: *n*-shot, one head-to-head per task vs. the cleaned pipeline output.

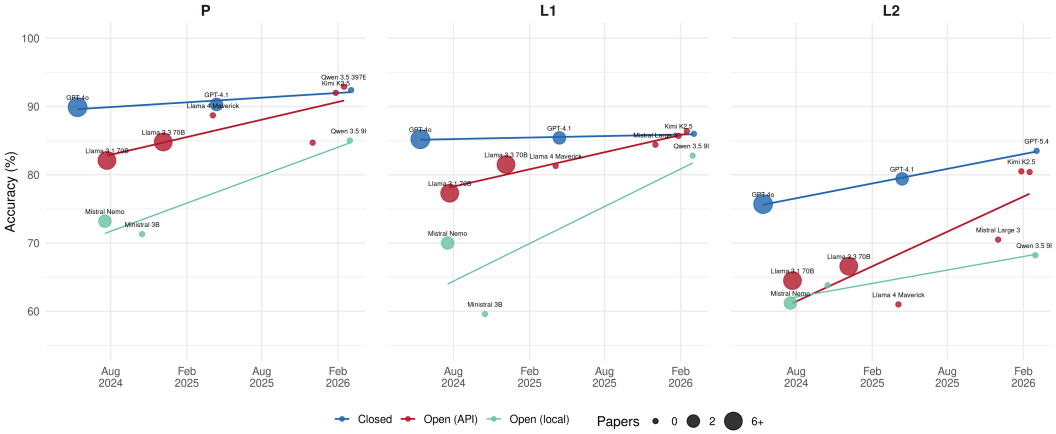
*Frontier robust to both. Smaller / open models: paraphrase rarely flips, raw codebook hurts.*

↗ per-task effects

# Result 6

- *Frontier models are insensitive to paraphrasing, verbosity, and noise.*
- *Raw codebook on the frontier: no significant difference from the cleaned version.*
- *Raw codebook on smaller / open models: significantly negative — the extra irrelevant content hurts.*

# Open or closed?




*Open has closed the accuracy gap, not the sensitivity gap.*

# Recommendations · for the researcher

*Provide a detailed codebook*, the same kind you would give a human annotator.

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*Large scale (100k+ items)* — label a subset with a frontier model, then distil into an open-weight model.

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*Formatting* (markdown, bullets) is fine — it helps *you* read and present the prompt, but makes little difference to a frontier model in terms of performance.

*Role persona* is fine if you want it. No reliable gain on frontier models, unpredictable on small / open models.

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*Read what the prompt contains*, not how it's packaged.

*Ask: are categories clearly defined?*

*Ask: are worked examples included?*

*Don't reward markdown or personas.*

*Don't demand paraphrase robustness checks on well-specified prompts.*

*Flag underspecified prompts on small / open models — expect instability there.*

Thank you.

can-celebi.github.io/prompting

Can Çelebi · can.celebi@univie.ac.at
Stefan P. Penczynski · s.penczynski@uea.ac.uk

*Much Ado about Prompting*

# Content · what drives accuracy

| | $P$ | $L_1$ | $L_2$ |
|---|---|---|---|
| *Classification instructions* | 2.6 % | **58.3 %** | 27.4 % |
| *Theory context* | — | 8.5 % | 9.6 % |
| *Experiment context* | 3.1 % | 1.3 % | 1.9 % |
| *Model choice* | **94.3 %** | 31.9 % | **61.1 %** |

*On L: classification definitions dominate. On P: model choice does.*

↗ per-model decomposition

↩ back

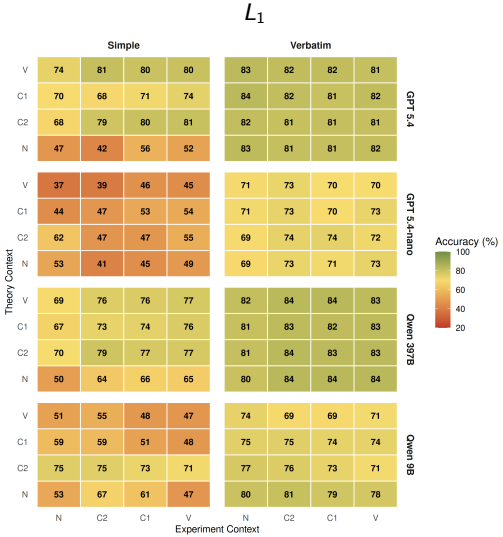
# Per-model variance decomposition

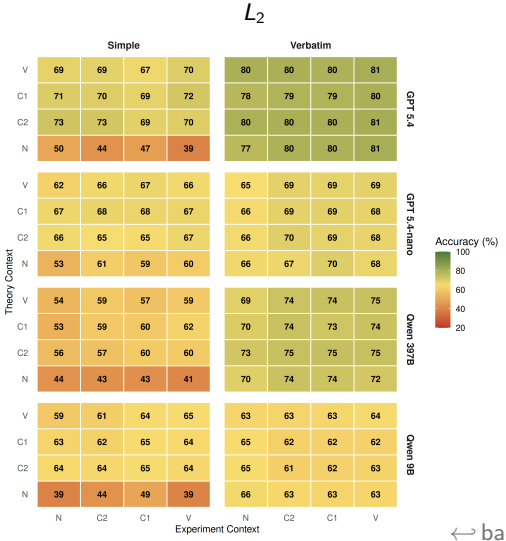
| | $P$ | | $L_1$ | | | $L_2$ | | |
|---|---|---|---|---|---|---|---|---|
| | Cls | Exp | Cls | Theory | Exp | Cls | Theory | Exp |
| GPT 5.4 | 20.6 | 79.4 | 57.4 | 41.6 | 0.9 | 69.7 | 30.2 | 0.1 |
| GPT 5.4-nano | 91.0 | 9.0 | 99.9 | 0.1 | 0.0 | 39.2 | 40.7 | 20.1 |
| Qwen 397B | 9.0 | 91.0 | 70.3 | 16.3 | 13.4 | 85.5 | 12.2 | 2.3 |
| Qwen 9B | 99.9 | 0.1 | 97.9 | 1.2 | 0.9 | 26.1 | 73.8 | 0.0 |

*Unique explained variance (%), Type-III LR tests.*

↩ back

# $L_1$ and $L_2$ surfaces · all four models



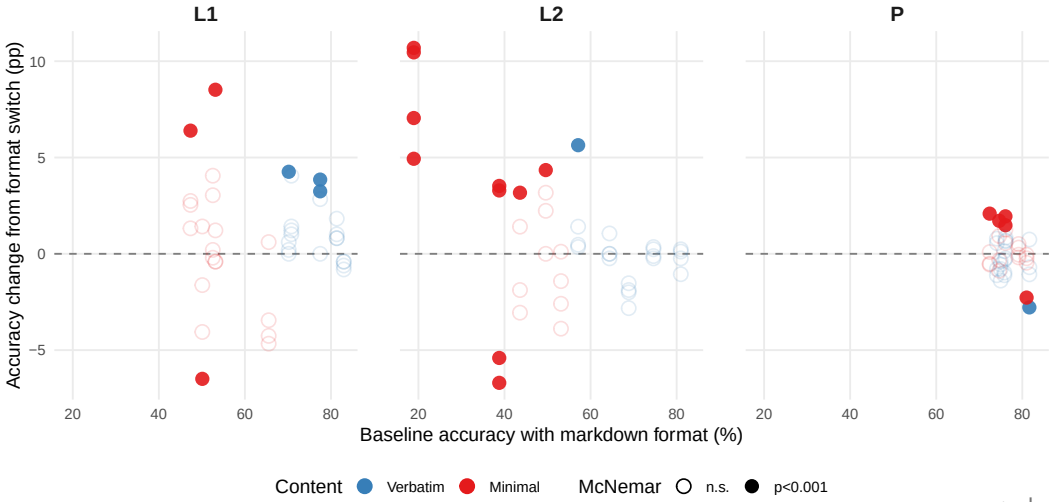

↩ back

# The five format variants

```
ml (markdown + list)                                                  baseline
## Level 0
- chooses randomly or without strategic reasoning

tl (titles + list)                                                     Level 0
- chooses randomly or without strategic reasoning

mp (titles + prose)                                                    Level 0
Chooses randomly or without strategic reasoning.

fl (flat list)                                                [header removed]
- chooses randomly or without strategic reasoning

pt (plain text)                                               [header removed]
Chooses randomly or without strategic reasoning.
```

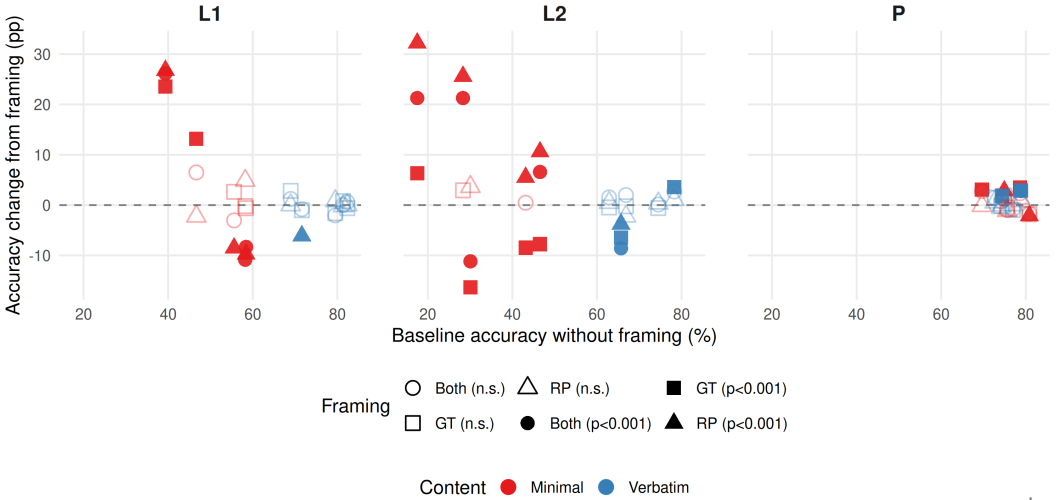
↩ back

# Format effects by task




↩ back

# Framing effects by task



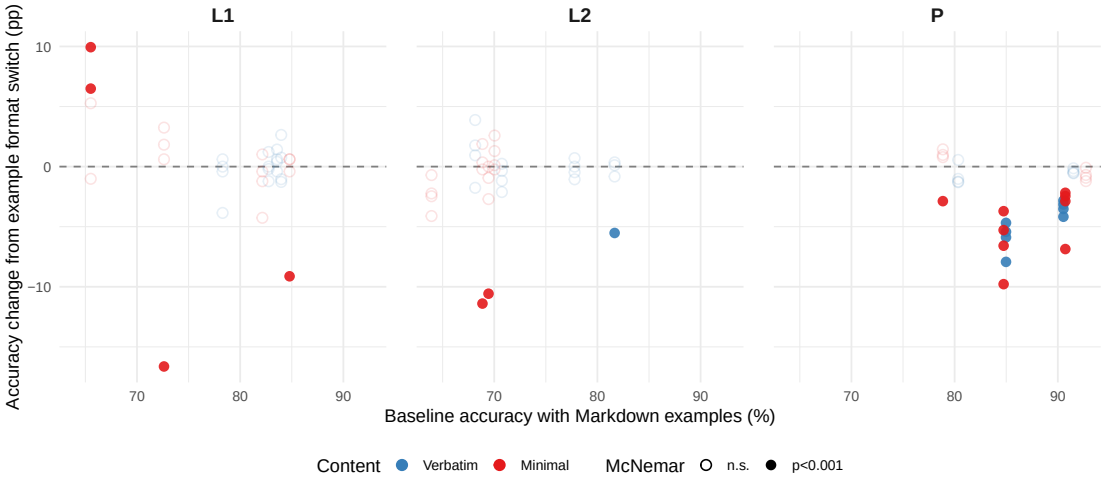

↩ back

# The five example-presentation styles

```
n (markdown)                                                    ## Example 1
### Message -- "P1:  all 200 then?  P2:  yes"
### Classification -- {"p1":"0", ...}

nx (XML)                <example> <message>...</message> ...</example>

nd (delimiter)            Example 1:  Message:  ... Classification:  {...}

na (arrow)                "P1:  all 200 then?  P2:  yes" → {"p1":"0",...}

t (multi-turn)  sent as user / assistant turns, outside the system message
```

↩ back

# Example format dispersion




↩ back

# Reasoning · effect at 0-shot vs. *n*-shot

| | *0-shot* | | | *n-shot* | | |
|---|---|---|---|---|---|---|
| | $P$ | $L_1$ | $L_2$ | $P$ | $L_1$ | $L_2$ |
| GPT 5.4 | −0.8 | +1.2 | 0.0 | +0.8 | +2.0 | +1.9 |
| GPT 5.4-nano | +4.9* | +5.5 | +1.6 | +8.3* | +1.0 | −0.4 |
| Qwen 397B | +5.9* | +1.0 | +0.6 | +2.5* | +2.8 | +2.6 |
| Qwen 9B | +8.6* | +10.5* | +2.4 | +6.5* | +1.6 | +4.0 |

*pp change vs. no reasoning. GPT: high effort; Qwen: on. * = significant.*

*Frontier: no gain. Smaller models: P-gains; L-gains nullified by examples.*

↗ effect by reasoning level ↩ back

# Reasoning — step-by-step levels (GPT models)

| | | 0-shot | | | *n*-shot | | |
|---|---|---|---|---|---|---|---|
| | Step | $P$ | $L_1$ | $L_2$ | $P$ | $L_1$ | $L_2$ |
| GPT 5.4 | off→low | +4.2* | +3.9* | +4.4* | +0.8* | +0.4 | −0.8 |
| | low→med | 0.0 | +1.0 | +2.4 | +0.4 | +2.0 | +1.9 |
| | med→high | −4.1* | −1.4 | −0.5 | 0.0 | −0.6 | +2.0 |
| | high→xhigh | −3.9* | 0.0 | +0.8 | −0.2 | −1.4 | −0.6 |
| GPT nano | off→low | +4.5* | +11.5* | +1.3* | +4.5* | +1.3 | −3.4* |
| | low→med | +0.9 | +5.1 | +0.2 | +3.0* | +0.6 | +3.4 |
| | med→high | 0.0 | −1.2 | +2.5 | +0.8 | −0.2 | +0.4 |
| | high→xhigh | −1.4 | +4.7 | +6.1* | +2.0* | +3.7 | +8.3* |

* = *McNemar* $p < 0.001$. *Each step relative to the previous level.*

↩ back

# Verbose vs. noise — an example

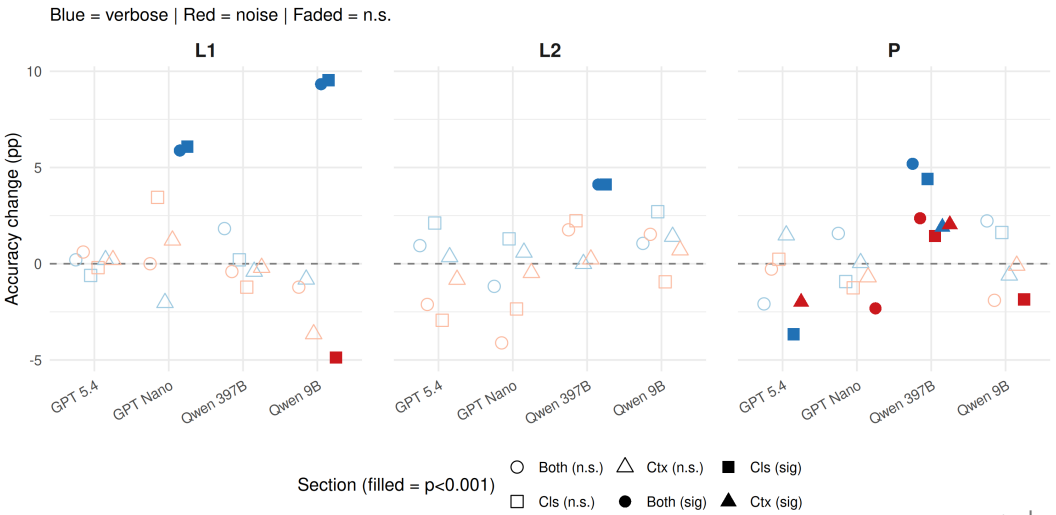
**Normal (verbatim)**
- Participants can invest between 0-200 in a group project.
- The total invested is doubled and shared equally.

**Verbose (relevant elaboration, +50-275% words)**
- Participants can invest between 0-200 in a group project, any integer.
- The total invested is doubled and shared equally. Investing 100 returns only 66 to the individual -- a net loss of 34, but the group gains 200.

**Noise (irrelevant padding, matched length)**
- Participants can invest between 0-200 in a group project.
- An adult red squirrel weighs 250-340 grams, body length 19-23 cm plus tail 15-20 cm.
- The total invested is doubled and shared equally.
- Red squirrels build nests called dreys, 6-12 m above ground, using twigs, moss, stripped bark.

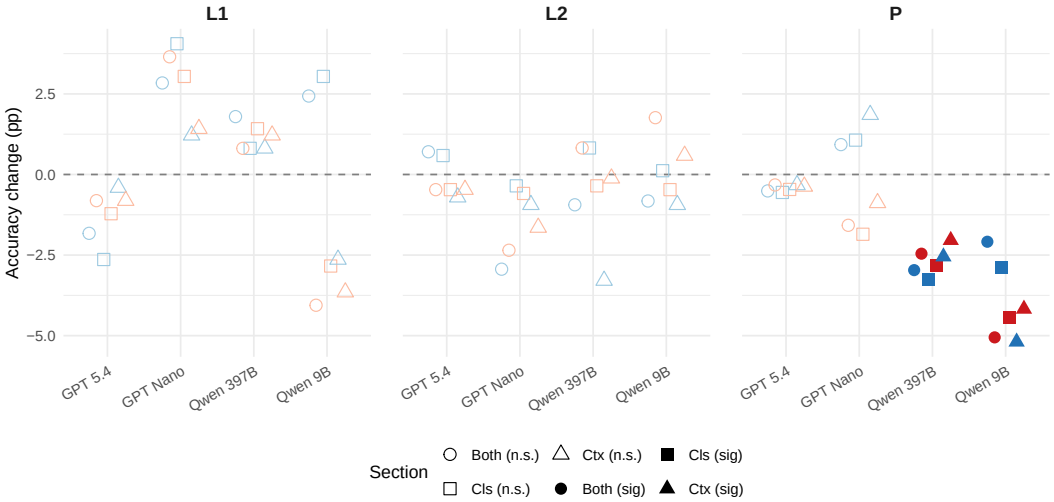
↩ back

# Verbose & noise (0-shot)




↩ back

# Verbose & noise (*n*-shot)



# Verbose & noise · mean $\bar{\Delta}$ and $\sigma$

| | *0-shot* | | | | *n-shot* | | | |
|---|---|---|---|---|---|---|---|---|
| | Verbose | | Noise | | Verbose | | Noise | |
| | $\bar{\Delta}$ | $\sigma$ | $\bar{\Delta}$ | $\sigma$ | $\bar{\Delta}$ | $\sigma$ | $\bar{\Delta}$ | $\sigma$ |
| GPT 5.4 | $-0.1$ | 1.8 | $-0.8$ | 1.2 | $-0.6$ | 1.1 | $-0.6$ | 0.3 |
| GPT 5.4-nano | $+1.4$ | 2.9 | $-0.8$ | 2.2 | $+0.9$ | 2.1 | $-0.2$ | 2.3 |
| Qwen 397B | $+2.3$ | 2.1 | $+0.9$ | 1.3 | $-0.9$ | 2.0 | $-0.4$ | 1.6 |
| Qwen 9B | $+3.1$ | 3.7 | $-1.1$ | 2.0 | $-0.9$ | 2.6 | $-2.5$ | 2.5 |

*In pp vs. verbatim baseline. 9 comparisons per cell.*

↩ back

# A note on scope

*Experimental-economics text is a best-case setting for codebook-based classification.*

- *Tightly constrained.* Fixed rules, payoffs, terms; edge cases rare. Unlike tweets, parliamentary speeches, Reddit threads.
- *Full context available.* Both *experiment context* and *formal theory* can be provided. Most social-science codebooks have neither.
- *Codebook shape differs elsewhere.* Political science / sociology: often many labels, short definitions, multi-label overlap.
- *First study* to vary experiment and theory context independently, and to cross them with formatting, framing, examples, reasoning.

Most directly relevant to *experimental economists*, then *economists using theory-based labels* for text classification.

Beyond that, generalisation needs care — a limitation of this study.

# Cosine similarity

The angle between two embedding vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$, ignoring their magnitudes:

$$\cos(\mathbf{a}, \mathbf{b}) \;=\; \frac{\mathbf{a}\cdot\mathbf{b}}{\|\mathbf{a}\|\,\|\mathbf{b}\|} \;=\; \frac{\sum_i a_i b_i}{\sqrt{\sum_i a_i^2}\,\sqrt{\sum_i b_i^2}}$$

- Range: $[-1, 1]$. *1* = same direction, *0* = orthogonal, *−1* = opposite.
- Measures *directional agreement*, not semantic equivalence — "I love it" and "I hate it" can still score high.
- Works well for *topical* / *structural* similarity; weaker on fine sentiment or stance.
- In our pipeline: cos $\geq 0.80$ between verbatim and any compressed variant (`text-embedding-3-large`).

↩ back

# Wilson confidence intervals

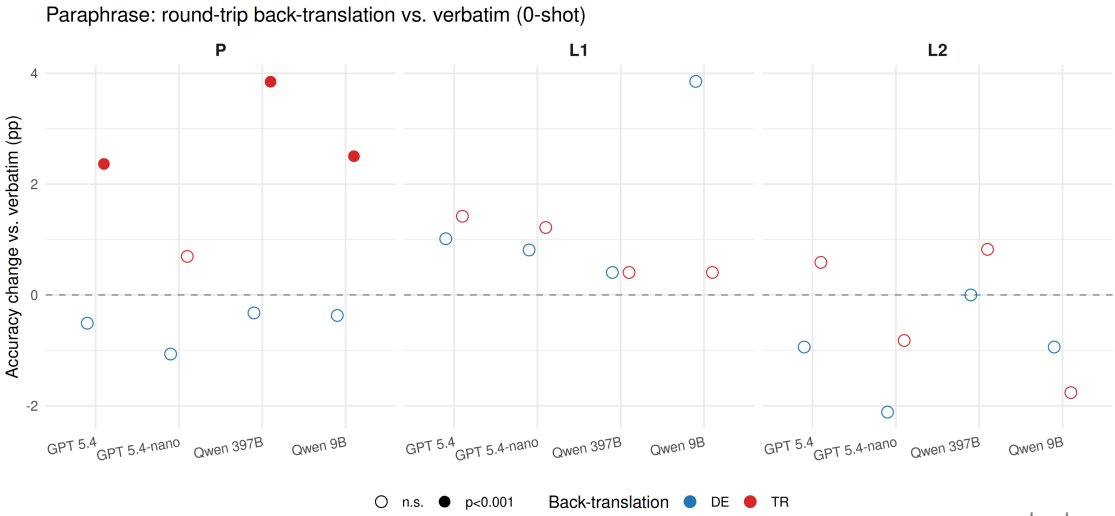
For a proportion $\hat{p} = x/n$ with $z = z_{1-\alpha/2}$:

$$\hat{p}_{\mathrm{W}} = \frac{\hat{p}+\frac{z^2}{2n}}{1+\frac{z^2}{n}} \pm \frac{z}{1+\frac{z^2}{n}}\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}$$

- Score-based; does not require $\hat{p}$ far from 0 or 1.
- Better coverage than the Wald (normal-approximation) interval when $n$ is small or $\hat{p}$ is near the boundary.
- Bounded inside $[0, 1]$ by construction.

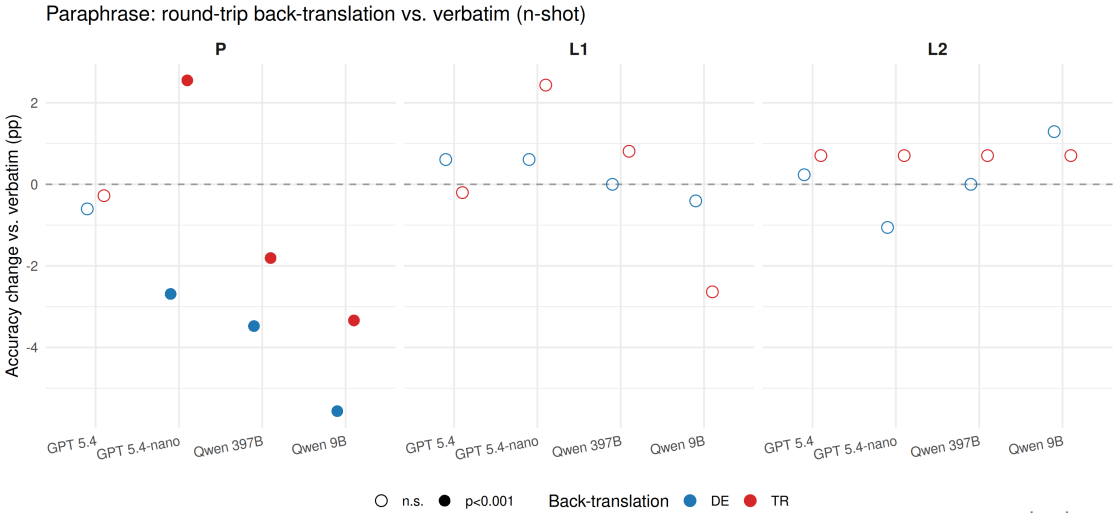
Wilson (1927), *JASA*.

# Paraphrase effects by model and task (0-shot)



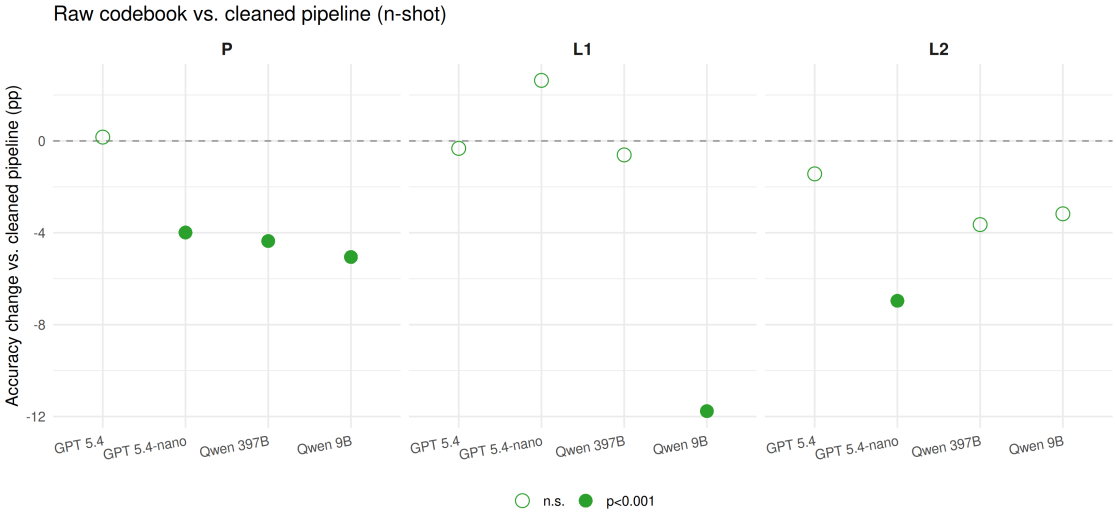

↩ back

# Paraphrase effects by model and task (*n*-shot)




↩ back

# Raw codebook effects by model and task




↩ back

# Where we land

*The prompt-engineering literature vs. what we actually find.*

| | *Prior claim* | *Our result* |
|---|---|---|
| Formatting | Sclar 24: +76 pp swings | well-specified: <1 pp; underspecified: scattered, no direction |
| Paraphrase | Reiß 23: flips on paraphrase | frontier: <1 pp; smaller models: mixed direction |
| Detail | Lupo 24: +21 pp; Rückdeschel 25: hurts small | helps large; too much context hurts small — both right |
| Persona | Törnberg 24; Dunivin 25: helps | well-specified: null; underspecified: mixed, no reliable gain |
| Reasoning | Dunivin 25: helps | frontier: no gain; small: helps recognition only, absorbed by examples |

*Prior claims hold in a specific corner — underspecified prompts on small / open models. On well-specified prompts and capable models the effects disappear.*

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